GUQINAGENT: Tool-Interactive Performance-Action Planning for Jianpu-to-Jianzipu Generation

Anonymous ACL submission

Abstract

Jianpu specifies musical pitches and rhythm, whereas jianzipu describes the actions used to perform them on the guqin. Generating jianzipu requires selecting pitch-correct, physically feasible actions that follow playing conventions and support the instrument's characteristic expression, or *yunwei*. These choices depend on tuning, inherited notation state, and the surrounding phrase. We introduce GUQINAGENT, a framework for generating jianzipu from complete jianpu scores through tool interaction. We provide a full-score benchmark and executable runtime, a two-stage agent that separates basic fingering from idiomatic refinement, and an expert-guided inverse trajectory construction method for training a student from annotated scores. Runtime tools expose pitch candidates, editable plans, and conservative warnings to support revision. Evaluation combines pitch correctness, structured action agreement, technique distributions, and task completion; performer assessment is planned to address qualities beyond automatic checks. On 48 scores, the two-stage full system achieves 99.981% pitch accuracy within 50 cents on 15,758 auditor-resolvable events. A single-stage student reaches 98.732% but has higher reference fingering and ornament agreement, revealing a trade-off between pitch consistency and reference action agreement. We will release the model, redacted training trajectories, benchmark, and runtime.

1 Introduction

A melody specifies which notes to play, but realizing it on the guqin also requires deciding how to produce and connect them. Jianzipu records these performance actions through combinations of strings, hui positions, left-hand fingers, right-hand strokes, and techniques (Li et al., 2026b,a). Preparing a jianzipu realization from a jianpu score therefore calls for instrument-specific judgment. A system that proposes editable realizations and exposes detectable errors could support performers and learners in arranging melodies for the guqin.

The goal involves four related requirements: pitch fidelity, physical playability, adherence to playing conventions, and idiomatic musical expression. Several string–position combinations may produce the same pitch, yet their hand shapes, transitions, and tonal effects can differ. Two individually valid notes may be difficult to execute together or connect naturally. Suitable choices of slides, vibrato, and compound gestures also contribute to *yunwei*, the guqin's characteristic expressive quality. Adding more ornaments alone does not establish this quality; their placement and relation to the phrase matter.

These requirements create three challenges. First, generation involves choosing among multiple realizations under both physical and musical constraints. Pitch agreement does not establish whether a hand position is feasible, whether a stroke follows conventional usage, or whether a technique suits the phrase. Second, decisions depend on the evolving state of the complete score. A harmonic region remains active until terminated, and subsequent notation may omit positions inherited from earlier events. The same string and hui can therefore imply different pitches inside and outside a harmonic region. Local edits must be interpreted in context and may change the meaning of later entries.

Third, expert annotations provide final realizations but generally omit the process of choosing, checking, and revising them. Final notation alone does not demonstrate when an agent should query pitch candidates, inspect preceding material, or respond to a warning. Recent jianzipu research addresses glyph recognition and structured extraction from manuscript pages (Li et al., 2026b,a). Here, the input is a complete machine-readable jianpu score, and the task is to construct a guqin perfor-

mance realization. This setting connects symbolic music generation with planning actions whose interpretation and feasibility depend on prior choices.

We address these challenges through interactive plan editing. Following work on reasoning with tools and iterative feedback (Yao et al., 2023; Schick et al., 2023; Madaan et al., 2023), GUQINAGENTpartitions a complete score into manageable work units while maintaining one shared notation plan. Its runtime provides tuning-aware candidates and conservative checks for pitch, state, and supported performance rules. A Fingering stage establishes a basic realization; a Guqinizer stage considers idiomatic technique and continuity. An offline teacher uses private expert annotations to construct actual tool interactions, which are redacted into student-visible supervision. Automatic checks support the verifiable parts of the task, while performer assessment is needed to judge expressive suitability. In summary, our contributions are fourfold:

1. We introduce *constrained jianzipu generation*, a benchmark and executable runtime for reproducible score-to-jianzipu evaluation.

2. We introduce warning-guided inference: deterministic pitch and conservative notation-constraint diagnostics become visible, actionable feedback rather than hidden post-hoc validation.

3. We design GUQINAGENT, a two-stage Fingering→Guqinizer agent that distinguishes basic sound production from stylistic refinement.

4. We develop targeted inverse trajectory construction, which replays annotations through the runtime and exports public, redacted supervision.

2 Related Work

Guqin notation and performance-oriented symbolic music. Recent work on jianzipu has focused on recovering structured notation from glyphs and manuscript pages (Li et al., 2026b,a). Earlier language-model-based work also studied transcription from jianzipu into modern notation (Shi et al., 2024). Our setting reverses this direction: starting from melodic events, the system must choose how to realize them on the instrument. A related problem arises in guitar tablature inference, where a pitch may admit several string–fret assignments and playability is assessed by musicians (Edwards et al., 2024). Work on piano fingering likewise models hand position and execution difficulty (Iman and Ahn, 2024). At the composition level, YuE2 generates a readable melody-and-chord score before synthesizing full-song audio; the score supports direct editing and revisions driven by external language models in response to user feedback (Yuan et al., 2026). GUQINAGENTstudies how a given score is realized through guqin performance actions, including state-dependent gestures and compact notation whose interpretation depends on tuning and prior actions.

Tool-interactive planning and revision. ReAct interleaves reasoning and environment actions, while Toolformer and ToolLLM investigate tool-use learning and data construction (Yao et al., 2023; Schick et al., 2023; Qin et al., 2024). Self-Refine and Reflexion use iterative feedback to improve model behavior; CRITIC specifically studies revision driven by external tool outputs (Madaan et al., 2023; Shinn et al., 2023; Gou et al., 2024). GUQINAGENTuses a shared executable notation state to check pitch realizations and issue conservative, event-level warnings. These checks inform subsequent edits but do not determine musical suitability on their own.

Learning from interaction trajectories. AgentTuning and FireAct demonstrate the value of supervised interaction trajectories for language agents (Zeng et al., 2024; Chen et al., 2023). More closely related, RetroGen reconstructs process supervision from final expert artifacts for evidence-grounded generation (Huang et al., 2026). Our teacher likewise has access to a target artifact during offline construction, but is required to execute edits in the student-facing guqin runtime; the retained training context contains only public actions and observations. GEPA instead optimizes prompts by reflecting on execution feedback without updating model weights, providing a distinct comparison to trajectory fine-tuning (Agrawal et al., 2026).

3 Methodology

3.1 Constrained Generation with an Executable Runtime

Given a complete jianpu score $x = (m, e_{1:N})$, the task is to construct an event-indexed jianzipu plan $y = (y_1, \ldots, y_N)$. The metadata m specifies the mode and tuning; each event e_i has a stable index, notated duration, structural type, and a target pitch

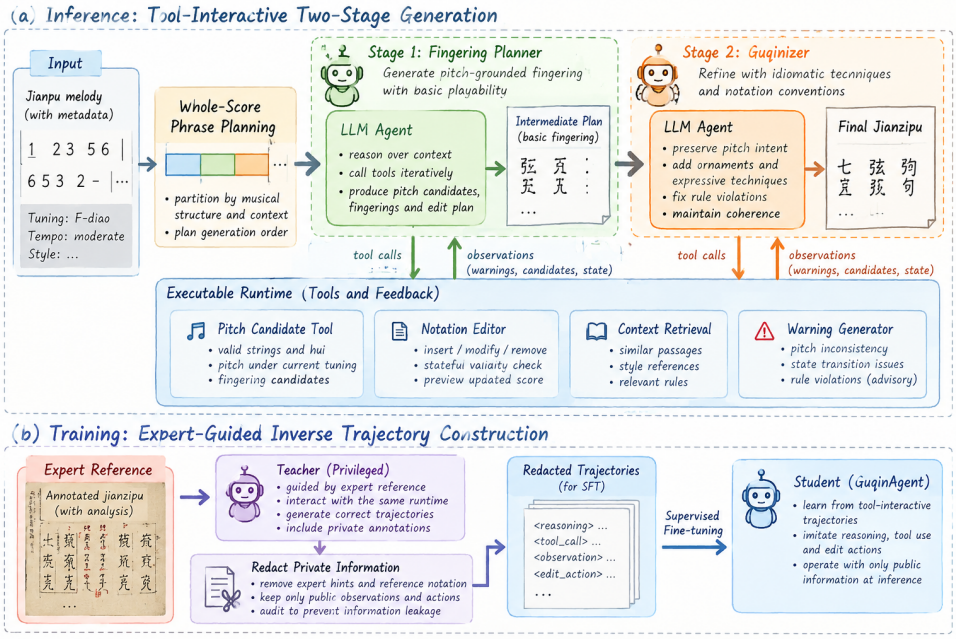


Figure 1: Overview of GUQINAGENT. **Top:** at inference time, the agent partitions a complete jianpu score into ordered work units, constructs a pitch-grounded intermediate plan in the Fingering stage, and refines it into final jianzipu in the Guqinizer stage. Both stages interact with the same executable runtime, which returns pitch candidates, plan previews, compact context, and advisory warnings. **Bottom:** for training, a privileged teacher uses expert annotations only through a private channel while interacting with the public runtime. Redaction retains a student-visible trajectory for supervised fine-tuning, so the deployed student conditions only on score context, its prior actions, and runtime observations.

q_i when defined. An output entry y_i is either a jianzipu expression or an explicit empty display value. The latter may indicate continuation or coverage by a compound gesture and does not remove the event from the score. ABC notation is used only as an internal representation for pitch processing, not as an additional input.

Unlike independent token prediction, the interpretation of y_i depends on preceding actions. We represent this dependence by a deterministic notation-state transition,

$$s_i = \mathcal{T}(s_{i-1}, y_i; m), \qquad s_0 = s_{\text{init}}(m), \quad (1)$$

where s_i records such properties as harmonic-region status, inherited left-hand position, walking gestures, and compound-action coverage. Given this state, the realized pitch $\hat{q}_i = \mathcal{P}(y_i, s_{i-1}; m)$ is a partial function: some compound or ambiguous expressions do not admit a reliable pitch estimate. Structural consistency and pitch agreement are therefore distinct from the broader question of musical suitability.

Interactive plan editing. The complete score is partitioned into an ordered sequence of bounded, musically aligned work units $\mathcal{B}(x) = (p_1, \ldots, p_K)$. This partition is computed from the input score; the benchmark examples remain complete scores. The agent edits one shared plan rather than generating independent phrases. At interaction step t within p_k, its public history consists of the score, current work unit, editable plan, replayed notation state, and previous tool observations:

$$\begin{aligned} h_t &= (x, p_k, y^{(t)}, s^{(t)}, o_{<t}), \\ a_t &\sim \pi_\theta(\cdot \mid h_t, r), \end{aligned} \quad (2)$$

where r denotes the current planning stage and $s^{(t)}$ is obtained by replaying the current plan. The executable runtime $\mathcal{R}$ processes an action and returns the resulting plan, state, and tool observation:

$$(y^{(t+1)}, s^{(t+1)}, o_t) = \mathcal{R}(x, y^{(t)}, s^{(t)}, a_t). \quad (3)$$

Read-only actions leave the plan unchanged. The runtime used to construct training trajectories is also used at inference time.

The agent can request tuning-aware string–hui realizations through `get_pitch_candidates`, including open strings, stopped notes, harmonics, and supported inherited or compound gestures. Returned candidates distinguish exact, approximate,

and unavailable matches. The only plan-writing operation is `edit_plan(jianzi_rows)`, which applies event-indexed updates and replays the resulting notation state. Compact context retrieval through `list_context` and targeted earlier-phrase lookup provides prior material without reproducing the entire score in every interaction.

Conservative constraint feedback. We distinguish runtime-checkable violations from cases whose interpretation is uncertain. For a rule c and event i, let $v_c(i; x, y, s) \in \{0, 1, \perp\}$ denote no detected violation, a detected violation, or an indeterminate check, respectively. The warning set is

$$W(x, y) = \{(i, c) : v_c(i; x, y, s) = 1\}, \quad (4)$$

with s obtained by replaying y. In the implementation, checks cover reliably parsed pitch mismatches and conservative diagnostics for harmonic-state continuity, walk direction or span, interrupted open-string resonance, paired stroke adjacency, and difficult left-hand reaches. Indeterminate cases are not converted into violations. Warnings accompany the relevant `edit_plan` preview but neither rewrite the plan nor imply that an unflagged output is musically correct. The agent may query candidates, revise the affected entries, or retain its choice; identical repeated edits are stopped to prevent unproductive loops.

3.2 Two-Stage Performance-Action Planning

The same notation event jointly specifies how a pitch is produced and how it is articulated. We separate these decisions into two successive planning stages. *Fingering* first constructs a basic pitch-grounded realization, choosing strings, hui positions, left-hand contact, and elementary right-hand strokes. *Guqinizer* then revises the committed plan for idiomatic execution, including ornaments, walking gestures, harmonic transitions, and compound-action notation. The second stage may preserve or revise earlier decisions; it is not a post-processing formatter.

For work unit p_k, let $y^{(k-1)}$ be the shared plan after completing the preceding units. A bounded runtime rollout under each stage gives

$$\begin{aligned} y^{F,(k)} &= \mathrm{Rollout}_F(x, p_k, y^{(k-1)}; \mathcal{R}), \\ y^{(k)} &= \mathrm{Rollout}_G(x, p_k, y^{F,(k)}; \mathcal{R}), \end{aligned} \quad (5)$$

where each rollout consists of stage-conditioned tool interactions of the form in Eq. (3). Only edits committed through the runtime enter the shared plan. Crucially, the state passed to p_{k+1} is replayed from $y^{(k)}$, including any earlier mistakes, rather than substituted from an expert reference. Both stages retain access to pitch candidates and runtime feedback. Warnings can trigger another revision but are not hard acceptance criteria; execution remains subject to the configured interaction budget.

This decomposition imposes a procedural bias: establish a basic sound-producing realization before making the more context-dependent choices involved in idiomatic execution. It does not assume a unique separation between fingering and ornamentation, nor does it guarantee that the second stage preserves every correct decision. We evaluate both the benefits and regressions introduced by refinement against a single-stage agent using the same public tools.

3.3 Expert-Guided Inverse Trajectory Learning

Expert jianzipu annotations specify desired performance actions but generally not the tool interactions by which an agent could arrive at them. Directly supervising final notation therefore leaves the interaction policy unspecified. We instead construct trajectories by allowing an offline teacher to work backward from the annotated target while acting through the same executable runtime available to the student.

Let z be a phrase-level expert annotation and $\mathcal{K}(z)$ the technique-relevant descriptions retrieved from a curated knowledge base. Both are private to the teacher. Its rollout is conditioned on the input, private guidance, and public runtime:

$$\tau^T \sim \mathrm{Rollout}\big(\pi_T \mid x, z, \mathcal{K}(z); \mathcal{R}\big), \quad (6)$$

where τ^T records the chronological sequence of tool actions, returned observations, and accepted plan updates. The reference guides the search for an executable realization but does not constitute a tool response. The teacher constructs stage-specific traces: basic fingering first, followed by idiomatic refinement. Semantic descriptions clarify technique constraints that may be implicit in compact jianzipu symbols; they are not added to the student's inference context.

Validation and public-prefix reconstruction. Candidate examples are screened for reference parsing and consistency before construction, and generated traces are checked against the runtime. Train-

ing examples are obtained by removing private expert content and retaining the public interaction sequence,

$$\tilde{\tau} = g_{\text{pub}}(\tau^T), \quad \mathcal{D} = \{\tilde{\tau} : \tau^T \text{ passes validation}\}. \tag{7}$$

The intended information boundary is temporal as well as structural: at each supervised action, the conditioning context should contain only the input, earlier public actions, and observations already returned by the runtime. Removing private fields alone cannot establish this property if rewritten reasoning anticipates future observations or reproduces unseen annotation details. Public-prefix consistency must therefore be checked separately from runtime execution validity; the precise checks and filtering counts belong in the experimental protocol.

Student supervision. We fit a stage-conditioned student policy to the assistant-side actions and messages in the retained public trajectories. For a supervised token position u, let h_u^{pub} contain the preceding student-visible context and let w_u be its target token. The training loss is standard masked cross-entropy,

$$\mathcal{L}_{\text{SFT}}(\theta) = -\sum_{\tilde{\tau}\in\mathcal{D}} \sum_{u\in\mathcal{Y}(\tilde{\tau})} \log p_\theta(w_u \mid h_u^{\text{pub}}), \tag{8}$$

where $\mathcal{Y}(\tilde{\tau})$ indexes supervised assistant tokens; tool outputs form context but are not prediction targets. At deployment, neither expert annotations nor private semantic guidance are provided. The resulting student must recover its actions from the input score, its own previous edits, and the executable feedback it has observed.

4 Experiments

4.1 Experimental Setup

Benchmark and Data Protocol. The test set contains 48 complete scores, partitioned into 1,031 work units, with 23,602 aligned event records. Scores are the units of dataset splitting, while phrases are internal work units used by the agent. The single-stage student and both two-stage SFT conditions produce protocol-valid outputs for all 1,031 work units.

All outputs are evaluated using the same current rules after replaying each complete score in chronological order. This preserves harmonic regions, left-hand positions, and other state across work-unit boundaries. Runtime snapshots and evaluation code fingerprints are retained for auditability. The final reproducibility record will specify the split manifest, checkpoint, training configuration, and inference arguments. [TODO: Add training-set size, adapter checkpoint hash, SFT hyperparameters, and inference budgets.]

Component	Contents
Input	Complete-score metadata, tuning, jianpu events, and stable event indices.
Output	One event-indexed jianzipu plan for the complete score.
Runtime	Pitch candidates, editing, context retrieval, validation, and diagnostic feedback.
Reference	Expert notation for offline supervision and structured action comparison.

Table 1: Benchmark representation. Reference notation is withheld from inference.

Evaluation Metrics. Pitch correctness is measured against the input jianpu melody. The expert jianzipu annotation supplies reference actions for fingering and technique metrics. Let $\mathcal{I}$ contain events for which the auditor returns a determinate matched or mismatched prediction. Let $a_i = 1$ if event i matches the target within 50 cents under the supported matching rules, and $a_i = 0$ otherwise. Let $\mathcal{A}_i$ contain the tone pairs aligned by the auditor, with absolute error δ_{ij} in cents for pair j. We report

$$\text{PitchAcc}_{50} = \frac{\sum_{i\in\mathcal{I}} a_i}{|\mathcal{I}|}, \quad \text{PitchMAE} = \frac{\sum_{i\in\mathcal{I}} \sum_{j\in\mathcal{A}_i} \delta_{ij}}{\sum_{i\in\mathcal{I}} |\mathcal{A}_i|}. \tag{9}$$

Accuracy uses events as its denominator; MAE averages aligned tone pairs, and a compound event can contribute multiple pairs. Unresolved cases are excluded from pitch scoring and must be reported separately. We report each condition's evaluable event count and accuracy on their intersection, since generated notation affects which expressions the auditor resolves. The complete event inventory includes structural and continuation events and is not the pitch denominator.

String F1 measures event-level string assignment against available reference labels. Left-hand fingering, right-hand fingering, and tone type are scored similarly and reported in Appendix A. Tone type distinguishes open, stopped, and harmonic notes. Ornament precision, recall, and F1 use event-level sets of parsed ornaments. These metrics measure

Method	Valid coverage ↑	Pitch Acc. ↑	MAE ↓	String F1 ↑	Tech. Sim. ↑	Orn. F1 ↑	Warn. Rate ↓
Original Qwen3.5-9B, direct	–	–	–	–	–	–	–
Original Qwen3.5-9B, agent†	1.07	N/A	N/A	N/A	N/A	N/A	N/A
GLM-5.3, ReAct + Handbook	100.00	99.760	2.542	36.514	0.6323	9.492	5.044
GLM API, agent + GEPA	–	–	–	–	–	–	–
Project SFT, single-stage agent	100.00	98.732	7.880	42.610	0.7618	24.622	14.258
Fixed-workflow SFT, single-stage	100.00	26.778	517.624	25.903	0.8401	31.446	95.829
Project SFT, two-stage, feedback OFF	100.00	86.237	64.993	38.554	0.7616	22.285	78.468
Project SFT, two-stage, feedback ON	100.00	99.981	2.308	39.761	0.7301	20.184	0.194

Table 2: Preliminary results on 48 scores. Coverage, accuracy, F1, and warning rate are percentages; MAE is in cents. Tech. Sim. is technique distribution similarity on $[0, 1]$. Valid coverage counts protocol-valid outputs over all 1,031 work units; warning rate counts work units with final runtime warnings. String F1 compares string assignments, not joint string–hui positions. †The original-model agent produced valid outputs for 11/59 attempted units; 972 were skipped after predecessor failures. N/A denotes unavailable full-test metrics; – denotes pending experiments. Detailed hand-fingering and tone-type metrics appear in Appendix A.

agreement with one expert realization; alternative playable realizations can receive lower scores. The current results do not include a joint string–hui F1.

Technique distribution similarity summarizes the relative use of named performance techniques. Each technique is counted at most once per event, then counts are normalized over all technique incidences in the corpus. For predicted and reference distributions p and q over their joint technique vocabulary $\mathcal{V}$, we use

$$\mathrm{TechSim}(p, q) = 1 - \frac{1}{2} \sum_{t \in \mathcal{V}} |p(t) - q(t)|. \quad (10)$$

The score lies in $[0, 1]$, with higher values indicating closer distributions. This metric excludes notation-state markers such as harmonic-region boundaries. It does not evaluate event order, and identical distributions can assign techniques to different events; ornament F1 retains a local comparison. Tone-type distribution similarity uses the same formula over open, stopped, and harmonic notes and is reported in the appendix.

Final warnings are counted by replaying completed plans through the runtime, including when warnings were hidden during inference. The rule-violation rate is the fraction of the 1,031 work units with at least one final warning; the total warning count is also reported. Unresolvable expressions do not automatically become violations, and a warning-free plan is not certified as playable.

Models and Comparisons. The feedback comparison uses Qwen3.5-9B with the same project-trained two-stage LoRA weights in both conditions. The full system exposes warnings, diagnostic hints, and warning-triggered follow-up interactions. The diagnostic-feedback-off condition hides this feedback and disables its follow-up mechanism while retaining structural validation and pitch candidates. Its public user and tool messages were checked for diagnostic-feedback leakage.

The single-stage student uses a separately trained Qwen3.5-9B LoRA and generates final jianzipu through the same tools. It is allowed up to 16 rounds per attempt, while the two-stage pipeline allows eight rounds per stage attempt and up to three Guqinizer passes. The nominal ceilings of 16 versus eight plus eight rounds do not equate aggregate inference budgets when retries and additional passes occur. We therefore compare the resulting pipelines without attributing every difference solely to stage decomposition. Both are evaluated on the same scores using complete model-generated histories and the same scoring rules.

We also evaluated the original Qwen3.5-9B checkpoint without the project LoRA using the two-stage agent and the same tool definitions. All 59 attempted inputs matched the test-input hashes. The original-model run allowed 8,192 output tokens per request, eight rounds per stage attempt, up to three attempts, and up to three Guqinizer passes, with a context limit of 16,384 tokens. After a work unit failed, the evaluator skipped the remainder of that score, preserving the requirement that history contain model-generated predecessors. These settings match the current SFT evaluation entry point; archived SFT outputs span multiple runs, so this comparison does not establish that every historical execution setting was identical.

The commercial-model ReAct + Handbook base-

line uses GLM-5.3 through its API with the production single-stage prompt and a fixed 73-entry technique handbook. It allows eight rounds and two attempts at temperature 0.25, uses the same tools and scoring pipeline, and produces protocol-valid outputs for all 1,031 units. A separate GLM API baseline with prompts optimized by GEPA on development data remains pending (Agrawal et al., 2026). These comparisons involve different model backends and do not isolate prompt optimization from model capability. GEPA's reflection model, API settings, optimization budget, and inference costs will be reported. Direct generation with the original Qwen checkpoint remains an optional additional comparison.

4.2 Main Results

Original-model completion and protocol failures. The original-model agent attempted 59 of 1,031 work units across the 48 scores. Eleven attempts produced protocol-valid outputs (18.64% of attempted units), while 48 failed. The remaining 972 units were skipped because a predecessor failed. Valid-output coverage was therefore 11/1,031 (1.07%); skipped units are excluded from the attempted-unit protocol-failure denominator. All three adapted conditions produced valid outputs for all 1,031 units. We do not report full-test musical metrics for the original model or compare its 11 successful units directly with the adapted model's complete-test scores.

Of the 48 failed units, 13 failed during Fingering and 35 during Guqinizer. Logs record output truncation in 45 failed units, at least one attempt reaching round eight in 26, and three consecutive pitch-candidate-query rounds in 31. These observations overlap and are not a mutually exclusive attribution of failures. They suggest that sustained tool interaction and termination behavior are substantial obstacles for the original model under the current protocol, alongside the musical decisions required by the task.

Technique distributions and the commercial-model baseline. Technique distribution similarity is 0.7618 for single-stage SFT, 0.7616 for two-stage feedback OFF, and 0.7301 for the two-stage full system. Thus, fewer runtime warnings do not imply closer aggregate technique usage. GLM-5.3 with ReAct + Handbook attains 99.760% pitch accuracy and 2.542 cents MAE, with 102 final warnings across 52 units (5.044%). Its technique similarity is 0.6323 and ornament F1 is 9.492, below both trained students. Its tone-type distribution similarity is higher, however (Appendix A). These measures capture different aspects of realization; the cross-model comparison alone does not establish that trajectory training is necessary for idiomatic playing.

Diagnostic	ON	OFF
Evaluable pitch events	15,758	15,360
Aligned tone pairs for MAE	16,335	15,958
Common evaluable events	14,289	14,289
Pitch Acc. on common events (%)	99.979	86.941
Final runtime warnings	2	3,436
Work units with warnings	2	809
Rule-violation rate (%)	0.194	78.468

Table 3: Diagnostic-feedback comparison. Counts refer to final plans, rescored under the same runtime with complete histories. Each condition covers 1,031 work units. OFF warnings are measured during evaluation despite being hidden during inference.

4.3 Ablation Studies

Pitch correctness and diagnostic feedback. The full system attains 99.981% pitch accuracy on 15,758 evaluable events, compared with 86.237% on 15,360 events when diagnostic feedback is disabled (Table 2). MAE is 2.308 versus 64.993 cents. Since the evaluable sets differ, Table 3 additionally reports accuracy on 14,289 events resolved in both conditions: 99.979% versus 86.941%, a gap of 13.038 percentage points. The observed accuracy gap therefore persists on the same input events. The result concerns auditor-supported realizations and does not establish accuracy on unresolved notation.

Residual warnings and error analysis. The full system leaves two final runtime warnings in two work units, compared with 3,436 warnings across 809 work units for OFF. The affected-work-unit rate is 0.194% versus 78.468%. One remaining ON warning in *Youlan* is an octave error, realizing E6 for a target E7. These cases warrant inspection of inherited state, compound notation, and ordinary pitch choices. Final-warning counts and pitch-error counts use different eligibility and matching rules and should not be treated as interchangeable metrics.

Fingering agreement and idiomatic realization. The full system's string, left-hand, right-hand, and tone-type F1 scores are 39.761, 57.119, 46.009, and 62.543, respectively. Their modest differences

Diagnostic	Single-stage	Two-stage
Evaluable pitch events	15,935	15,758
Common evaluable events	14,580	14,580
Common-event Pitch Acc. (%)	98.807	99.979
Final runtime warnings	381	2
Work units with warnings	147	2
Rule-violation rate (%)	14.258	0.194

Table 4: Single-stage and two-stage full-system comparison on the same 48 scores, scored after replaying complete histories. Both achieve 100% valid-output coverage. Different retry and pass structures preclude interpreting this as a strictly equal-total-budget architecture ablation.

Diagnostic	Fixed	Teacher
Evaluable pitch events	13,556	15,935
Common evaluable events	12,792	12,792
Common-event Pitch Acc. (%)	26.931	98.781
Final runtime warnings	11,596	381
Work units with warnings	988	147
Warning rate (%)	95.829	14.258

Table 5: Single-stage trajectory-construction comparison. Fixed uses a scripted query-and-edit sequence; Teacher uses expert-guided interactive trajectories. Both produce valid outputs for all 1,031 units. Some training prompts also differ between the recipes.

from OFF accompany a much larger pitch-accuracy gap. The full system also has lower ornament F1: 20.184 versus 22.285. The current comparison provides evidence of improved pitch consistency without establishing an advantage in ornament selection.

Tone distributions indicate that pitch accuracy leaves room for different realization strategies. Among recognized tone types on reference-labeled events, harmonics account for 7.51% of ON predictions, compared with 29.34% of reference tone types. A system can realize the correct pitch through different strings, positions, or tone types, and can also underuse stylistically appropriate techniques. Both possibilities motivate examining action distributions and consulting performers when assessing idiomatic quality.

Single-stage versus two-stage planning. The single-stage student completes all 1,031 work units, with 98.732% pitch accuracy on 15,935 evaluable events and 7.880 cents MAE. The two-stage full system reaches 99.981% on 15,758 events and 2.308 cents MAE. On their 14,580 common evaluable events, accuracy is 98.807% for single-stage and 99.979% for two-stage, a gap of approximately 1.173 percentage points. Table 4 reports denominators and final warnings for this comparison.

The single-stage student has higher reference agreement in every reported action category: string F1 is 42.610 versus 39.761, left-hand F1 is 57.917 versus 57.119, right-hand F1 is 47.697 versus 46.009, tone-type F1 is 65.629 versus 62.543, and ornament F1 is 24.622 versus 20.184 (Table 2). The two-stage pipeline has fewer final warnings, 2 versus 381, with warning-bearing work units accounting for 0.194% versus 14.258%. These results show a trade-off: the current two-stage pipeline achieves greater pitch and runtime-rule consistency, while single-stage generation more closely reproduces the reference actions and ornaments. Performer assessment is needed to determine how these differences affect execution and musical expression.

Teacher trajectory construction. We compare the standard single-stage student with a student trained on fixed-workflow trajectories. Each fixed trajectory contains generic reasoning, one real pitch-candidate query, and one edit constructed from the expert annotation, without a teacher API rollout. Both students are evaluated with diagnostic feedback and a 16-round stage limit on the same 48 scores using complete generated histories. This comparison keeps the architecture single-stage; training trajectories and some system/user prompts differ, so it evaluates the construction recipes rather than an isolated causal effect of interactive teacher reasoning.

The fixed-workflow student achieves 100% protocol-valid coverage, but pitch accuracy is 26.778% on 13,556 evaluable events and MAE is 517.624 cents. The standard single-stage student attains 98.732% on 15,935 events and 7.880 cents MAE. On their 12,792 common evaluable events, accuracy is 26.931% versus 98.781% (Table 5). Thus, the large pitch gap persists on a shared event set rather than arising solely from different parseability.

The fixed-workflow model leaves 11,596 warnings across 988 units, including 9,926 pitch-mismatch warnings. Its string F1 is 25.903 versus 42.610 for the standard single-stage student. However, its ornament F1 is higher (31.446 versus 24.622), as is technique distribution similarity (0.8401 versus 0.7618). Matching reference technique usage therefore does not ensure that those actions realize the input pitches. The results fa-

vor expert-guided trajectories for pitch consistency under the evaluated recipes, while leaving the separate roles of reasoning content, iterative feedback, prompt differences, and training length unresolved.

4.4 Expert Evaluation

A blinded performer study will complement automatic metrics. Proposed dimensions are physical execution feasibility, conformity to playing conventions and continuity of hand movements, and expressive suitability or *yunwei*. The last dimension concerns how slides, vibrato, and compound gestures serve the musical phrase; it is not defined by ornament count or reference F1. Examples will be anonymized and presented in randomized order, with input scores available to raters. The study will report rater experience, sampling, agreement between raters, and uncertainty of the estimates. Expert scores and claims of superior playability or expression await this assessment.

5 Conclusion

GUQINAGENTframes jianzipu generation as executable, stateful plan editing. Its benchmark makes score events, tuning, and notation state explicit; its two-stage architecture separates basic realization from idiomatic refinement; and its inverse trajectories create public tool-use supervision from private annotations. On 48 scores, both trained students complete all 1,031 work units. The two-stage full system attains higher auditor-resolvable pitch accuracy and fewer final warnings, while the single-stage student achieves higher reference agreement across fingering, tone type, and ornaments. These complementary outcomes motivate evaluating execution quality and musical expression alongside pitch consistency. The fixed-workflow trajectory baseline attains closer aggregate technique usage but much lower pitch accuracy than the standard single-stage student, underscoring the distinction between reproducing action patterns and realizing the input melody. The GLM-based GEPA comparison and performer assessment remain to be completed.

6 Limitations, Ethics, and Resource Availability

The auditor is conservative and incomplete: no warning is not proof of musical correctness, and unresolved notation is not a success. Fine-grained left-hand reachability, timbre, and lineage-specific style still require expert review or additional physical modeling. The digitized score collection may not represent all historical sources or playing traditions. We position GUQINAGENTas an assistive research system, not a replacement for performers or tradition bearers. Before release, we will review source licenses and provenance, remove private teacher material, and publish only redacted trajectories intended for public supervision. Since inference feedback and final warning scoring share domain rules, warning reductions are not independent evidence of performer-rated quality. Model-dependent parseability changes the pitch denominator even with a fixed input score. The stage comparison uses different retry and pass structures, so nominal round ceilings do not establish equal total inference budgets. The original-model agent's low output coverage also prevents a full-test musical-quality comparison. The trajectory-construction comparison also includes differences in training prompts and trajectory lengths, limiting causal attribution to individual construction steps. GEPA and expert evaluation remain pending.

Open Science and Resource Availability. We will release the trained model, redacted training trajectories, the test benchmark, and the executable runtime used by teacher generation and inference. The release will include score-level split definitions, event-indexed schemas, evaluation scripts, filtering manifests, and instructions for reproducing the benchmark. Private teacher prompts, unredacted teacher reasoning, and any source material that cannot be redistributed will not be included. The archival repository URL, model license, dataset license, and versioned release identifier will be added before submission: [TODO: repository URL and licenses].

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A Additional Action and Tone-Type Results

Table 6 reports event-level left- and right-hand fingering and tone-type F1, alongside tone-type distribution similarity. These complement the pitch, string-assignment, technique-distribution, ornament, and warning metrics in the main comparison. Tone similarity compares aggregate proportions of open, stopped, and harmonic notes and does not require these tone types to occur at the same events.

Method	LH F1	RH F1	Tone F1	Tone Sim.
GLM + Handbook	43.609	40.835	54.555	0.8439
Single-stage SFT	57.917	47.697	65.629	0.8113
Fixed-workflow SFT	54.965	43.505	59.102	0.8320
Two-stage OFF	56.599	45.659	61.670	0.7810
Two-stage ON	57.119	46.009	62.543	0.7743

Table 6: Additional structured action metrics on the same 48-score test set. F1 is a percentage; tone similarity is $1 - \mathrm{TVD}$ on $[0, 1]$. Higher values indicate closer agreement with reference annotations. All five conditions produce protocol-valid outputs for all 1,031 work units.